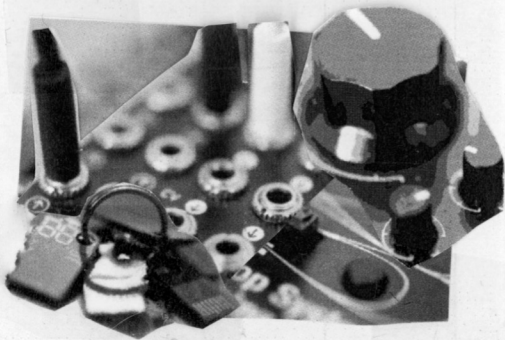


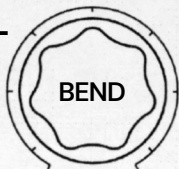
BENDS



45

CIRCUIT BENT
EFFECTS ENGINE

/ PANEL



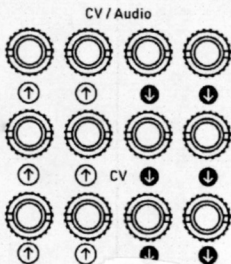
WIDTH
ROUTE



FREEZE
PERFORM
PAGE



IN L
IN R



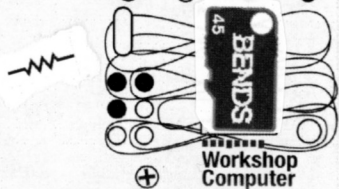
BEND
GLITCH

CLOCK
FREEZE

OUT L
OUT R

PITCH
RANDOM

TRIGGER
TEXTURE



/ NAVIGATION

TAP DOWN Cycles through the pages for each effect (indicated by LEDs)

HOLD DOWN Adjust Global settings while switch is held down. If nothing is adjusted, it saves the current state after 3 seconds.

HOLD DOWN + RESET BUTTON
Reset Bends to defaults

/ GLOBAL

MAIN Bend Macro
X Stereo Width
Y Effects Routing



Main knob adjusts all degradation parameters at once.

Y Knob effects chain:
1.CLDGFR 2.RFCDGL 3.GLDCFR
4.FLCDGR 5.RESET

/ CHORUS

MAIN Chorus Mix 
X LFO Speed/ Pitch 
Y Chorus Depth/ Feedback 

Y at high values (>3 o'clock) turns chorus into a Karplus-Strong string resonator.

If this mode is active, the X-Knob + CV1 set the resonator pitch.

/ LOSS

MAIN Loss Mix 
X Crush Type 
Y Jitter Type 

X blends between Tape, MP3 Ringing, Bitcrush, and Decimation loss.

Y blends between Tape Wow/Flutter, Packet Dropouts, and CD Sputter / random XOR glitches.).

/ DELAY

MAIN Delay Mix 
X Delay Time 
Y Feedback 

When a clock is present, delay time snaps to divisions of the clock.
Short times + feedback = resonator

Longer delay times reduce delay clock speed, leading to nice aliasing

/ GLITCH

MAIN Glitch Probability 
X Glitch Length 
Y Glitch Type 

When a clock is present, glitch length snaps to divisions of the clock.

Y drives glitch type: stutter / stutter + reverse / stutter + reverse + random pitch.

/ FILTER

MAIN	Filter Cutoff (Low/ High)	☐ ☐
X	Filter Resonance	☐ ☐
Y	Drive/ Wavefolder	☒ ☐

Morphing state-variable filter driven by a wavefolder. Turning the Main knob left selects low-pass; right selects high-pass. At high resonance settings, the filter self-oscillates (CV1 acts as 1V/Oct pitch input).

/ REVERB

MAIN	Reverb Mix	☐ ☐
X	Size/ Type (Spring/ Hall)	☐ ☐
Y	Damping	☐ ☒

The X knob blends from spring (left) to hall (right) reverb while adjusting room size.

Higher damping leads to bitcrushing in the reverb feedback loop.

/ FREEZE

MAIN	Block Position	☐ ☒
X	Block Length	☒ ☐
Y	Pitch Shift	☐ ☒

Flicking the switch up freezes the buffer. The Main knob controls buffer position (left: LFO; right: random).

When a clock is present, freeze block length and position snap to clock.

/ OUTPUTS

CV Out 1:
Clocked Turing Machine sequence (controlled by Glitch Page Y-knob).

CV Out 2: Stepped Random S&H

Pulse 2 / No Audio Inputs:
When no input is connected, the module self-generates lo-fi micro-glitches driven by the Loss Page.

/ BENDS

Bends is a 6-stage stereo multi-effects processor, micro looper, and signal degradation engine designed for the Music Thing Workshop Computer.

The card makes imperfection its core concept. Inspired by 90s circuit-bent rack units, Bends exposes the artifacts, memory corruption, and glitches of early digital processors as an effect, but also as an instrument.

Create ambient drones, subtle distortions, or glitchy drum loops, or sequence a distorted industrial string lead—it's your choice!

/ Download

[computer.musicthing.co.uk/
programs/45-bends/](http://computer.musicthing.co.uk/programs/45-bends/)

/ Vincent Maurer 2026

vincentmaurer.de

